## Copilot

Certainly! The **Sieve of Eratosthenes** is an ancient and efficient algorithm for finding all prime numbers up to a given limit. Let's break it down:

1. **Objective**: To find all prime numbers less than or equal to a given integer `n`.
2. **Steps**:
   - Create a list of consecutive integers from **2 through `n`**.
   - Start with the smallest prime number, which is **2**.
   - Enumerate the multiples of this prime by counting in increments of `p` from **2p** up to `n`, and mark them as composite (not prime).
   - Find the smallest unmarked number greater than `p` (if it exists). Let this new number be the next prime, and repeat the process from step 3.
   - Continue until there are no more unmarked numbers greater than `p`.
3. **Termination**:
   - The remaining unmarked numbers in the list are all the primes below `n`.
4. **Refinements**:
   - Initially, list only **odd numbers** (3, 5, …, `n`) to reduce memory usage.
   - Mark multiples starting from `p^2` (since smaller multiples have already been marked).
   - The algorithm can terminate when `p^2` exceeds `n`.

Remember, every value assigned to `p` during the process will be prime, as any composite number would have been marked as a multiple of some smaller prime. Some numbers may be marked more than once (e.g., 15 is marked both for 3 and 5).

The Sieve of Eratosthenes is a powerful tool for generating prime numbers efficiently! 🌟 🔢

For more details, you can explore the Wikipedia article or other resources . Happy coding! 👩‍💻 🚀